Journal of Molecular Structure 1273 (2023) 134283

Contents lists available at ScienceDirect

Journal of Molecular Structure

journal homepage: www.elsevier.com/locate/molstr

Structure–properties relationships of deep eutectic solvents formed between choline chloride and carboxylic acids: Experimental and computational study

Man Zhang[a,b], Xiaomeng Zhao[a], Siyang Tang[a], Kejing Wu[b], Binshen Wang[b], Yingying Liu[b], Yingming Zhu[b], Houfang Lu[a,b,*], Bin Liang[a,b]

[a] School of Chemical Engineering, Sichuan University, Chengdu 610065, China
[b] Institute of New Energy and Low-carbon Technology, Sichuan University, Chengdu 610207, China

ARTICLE INFO

Article history:
Received 16 August 2022
Revised 17 September 2022
Accepted 4 October 2022
Available online 5 October 2022

Keywords:
Deep eutectic solvent
Carboxylic acids
Physicochemical properties
Interaction
Density functional theory

ABSTRACT

To implement different applications of deep eutectic solvents (DESs), the knowledge of their structure and properties should be enhanced. In this work, experimental characterizations together with theoretical calculations were employed to study the structure–properties relationships of seven ChCl-carboxylic acids DESs. The obtained DESs were analyzed using NMR spectroscopy to check their structures and interactions. The physicochemical properties of DESs, such as viscosity, conductivity, and Kamlet-Taft solvatochromic parameters, are closely related to the structure of carboxylic acids. A DES with more hydroxyl or carboxyl groups has a greater viscosity and a lower conductivity. Increasing the carbon chain of carboxylic acid reduces the polarity of the corresponding DES. Moreover, the calculation results from density functional theory (DFT) indicate that the type and intensity of intermolecular interaction of DESs are closely related to the structure of the carboxylic acid, which are consistent with experimental results. The systematic research on the structure–properties relationships expand our knowledge on the rational design of task-specific DESs.

© 2022 Elsevier B.V. All rights reserved.

1. Introduction

Currently, deep eutectic solvents (DESs) have received great attention as one of the most promising alternatives to ionic liquids (ILs) and conventional organic solvents [1]. A DES is a mixture of hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD), and the eutectic point temperature is below that of an ideal liquid mixture presenting significant negative deviations from ideality [2]. DESs can be easily prepared from some natural, renewable, and low-cost chemicals, such as carbohydrates, polyols, amino acids, and organic acids [3]. From the perspective of green chemistry, DESs are considered as harmless solvents due to their biodegradability, recyclability, and reusability [4].

Studying the physicochemical characterization of DESs is important and essential before reaching the level of their industrial use [5]. Many efforts have been devoted to the physicochemical property data of DESs. Recently, Fernando et al. reported the viscosities and conductivities of glucose: choline chloride: water (GCH) and glucose:citric acid:water (GCiH) DESs, which were 1193 mPa·s, 0.741 mS·cm^{-1} and 491 mPa·s, 0.081 mS·cm^{-1}, respectively [6].They found that the solubility of quercetin in the GCH was higher than in GCiH, which may be due to the stronger supramolecular structure formed by GCH components. Dai et al. used various DESs, such as lactic acid-glucose, proline-malic acid, ChCl-sucrose, ChCl-propanediol, to extract phenolic compounds from safflower. Their results showed that the extraction efficiency was related to the viscosity and polarity of DESs [7], among which ChCl-sucrose was the most viscous, resulting the poor extraction efficiency of compounds from biomass. ChCl-propanediol had the lowest polarity among all the tested DESs and showed the lowest extraction efficiency for polar compounds. And it was proved that both the polarity (π^*) and basicity (β) of DESs played an important role in the lignin extraction [8]. A DES with higher π^* and β values can dissolve lignin more easily. However, the catalog of all potential DESs covers a huge proportion of the library of known compounds. Studies concerning the physicochemical properties of DESs are still incomplete.

Similar to ILs, DESs have emerged as designable solvents for many applications [9]. Changes in the structure of HBA and/or HBD can affect the interactions and thus modify the properties of DESs [10,11]. Polycarboxylic acids based DESs have higher viscosity than

* Corresponding author.
E-mail address: luhouf@163.com (H. Lu).

https://doi.org/10.1016/j.molstruc.2022.134283
0022-2860/© 2022 Elsevier B.V. All rights reserved.

polyols or phenols based DESs due to the stronger interaction between HBA and HBD [12]. By using density functional theory (DFT) calculations, Gautam et al. confirmed the existence of hydrogen bonds in the structures of ChCl-acetic acid and ChCl-formic acid DESs [13]. Dwamena et al. demonstrated that both solvatochromic molecular probes and DFT calculations were effective methods for studying the polarity parameters and charge transfer of DESs [14]. They found that the polarizability (π^*) and hydrogen bond acidity (α) decreased with increasing alkyl chain length. However, up to now, the understanding of the relationship between the structures of carboxyl and hydroxyl groups in HBD and the physicochemical properties of the corresponding DESs remains in a relatively infantile state.

There is a growing interest in those DESs composed of carboxylic acids that act not only as solvents but as active liquid ingredients, i.e., as precursors and building blocks for final materials, catalysts and structure-directing agents, etc [15]. An in-depth understanding on the structures and properties of this kind of DESs has significant implications for their various applications. Herein, this work focuses on describing the relationships between the structures and properties of DESs formed by ChCl and carboxylic acids. Based on our previous studies on the preparation strategy of DESs, seven carboxylic acids with good thermal and physical stability were selected [16]. Nuclear magnetic resonance (NMR) techniques were used to validate the molecular structure of the HBA and HBD. We presented the viscosity, conductivity, and Kamlet-Taft solvatochromic parameters of seven carboxylic acid-based DESs and discussed the influence of functional groups on the physicochemical properties based on the molecular structure of acids. In order to rationalize the intermolecular interactions generated among the components, theoretical studies were performed by using DFT computational method. The work can provide further insight into the structure and properties of DESs, which may provide useful information for designing DESs and predicting their potential applications.

2. Materials & methods

2.1. Materials

ChCl (purity ≥ 99%) was obtained from Chengdu Chron Chemical Reagent Co., Ltd., China. Oxalic acid (Oxa, ≥ 98%), malonic acid (Malo, ≥ 99%), glutaric acid (Glu, ≥ 99%), DL-malic acid (Mali, ≥ 99%), α-ketoglutaric acid (Ket, ≥ 98%), and citric acid (Cit, ≥ 99%) were purchased from Aladdin Reagent Co. Ltd., Shanghai, China (Table 1). DL-lactic acid was acquired from TCl Shanghai Co. Ltd., China. with a purity of 85.0%. The molecular probes of the Nile red (≥ 99% mass fraction purity) and 4-nitroaniline (≥ 98% mass fraction purity) were all purchased from Aladdin Industrial Inc. N, N-Diethyl-4-nitroaniline (≥ 98% mass fraction purity) was provided by J&K Scientific Ltd. Ultrapure grade water used in the experiments was obtained by a Milli Q water purification system (Res. J scientific instruments CO., LTD, Xiamen, China). All other chemicals were of analytical grade.

2.2. Methods

2.2.1. Preparation of the DESs

DESs were obtained by mixing ChCl with different carboxylic acids. Based on our previous research, binary mixtures with the molar ratio of 1:1 were magnetically stirred and heated [16]. ChCl-Cit DES was prepared at 120°C for 12 h, while the other DESs were obtained by stirring at 80°C for 1 h or more to form a clear liquid. Once the homogeneous and transparent liquid was formed, the mixture was cooled down to room temperature in a desiccator to avoid moisture absorption. The raw materials were all dried in a vacuum oven and placed in a moisture controlled area. All prepared DESs were vacuum dried at 60°C for approximately 48 h and stored in a vacuum desiccator before physical property analysis.

2.2.2. Characterization of the DESs

All the NMR experiments were carried out on a JNM-ECZ400S spectrometer (JEOL Ltd. Japan) with 400 MHz at room temperature. Eight scans were recorded with a standard one-pulse sequence for each ^{1}H NMR spectrum, and 512 scans were collected for each ^{13}C NMR. Intermolecular interactions between the components of ChCl−Glu (1:1, mol/mol) could be distinguished using ^{1}H–^{1}H nuclear overhauser effect spectroscopy (NOESY). A mixing time of 400 ms, a pulse delay of 2.0 s and 16 scans for each increment were recorded. The DMSO-d6 (0.6 mL) was used to dissolve all the samples (30 mg) while tetramethylsilane (TMS) was used as an internal reference.

2.2.3. Viscosity and electrical conductivity measurements

The viscosities and electrical conductivities were obtained with a cone-plate viscometer (Brookfield, DV2TRVCP) and conductivity instrument (DDS801, Guiyang Xuetong instrument, and Meter Co., Ltd. China), respectively. The viscometer was calibrated at 298.1 K by a standard sample (silicone oil) supplied by Brookfield, which is Newton fluid and the viscosity is 858 mPa·s. And the relative uncertainty is smaller than 2%. The electrode constant of the conductivity meter (k) was calibrated using a standard KCl solution. For comparison purpose, we measured these properties for methanol and glycerol, the relative uncertainty is ± 2% at local atmospheric pressure (96 kPa ± 1 kPa). Three sets of measurements were obtained for each DES at given temperatures using a water bath, then the mean value of these repeated measurements was reported.

2.2.4. Solvatochromic probes

As for the measurement of Kamlet-Taft polarity parameters of DESs, N, N-Diethyl-4-nitroaniline was used as a dye indicator for determining the value of π^*, which reflected the polarizability of the solvent [17]. Besides, the Nile red and 4-nitroaniline were respectively used as solvatochromic probes to measure the values of α and β, which represented hydrogen bond acidity and hydrogen bond alkalinity, respectively [18]. A stock solution of 1 mM dye in methanol was prepared. Then, an appropriate amount of the dye stock solution was micropipetted and transferred to the clean and dry glass vials to mix with different DESs. The dye-DES mixtures (4 mL) were then dried at 60°C under a vacuum for 48 h to remove methanol. The absorbance was less than 0.8 a.u. via a scan of the UV–vis spectra using an Optizen 3220 UV spectrophotometer. All spectroscopic measurements were performed in triplicate. The solvatochromic parameters were calculated using the following equations [19]:

$$E_T(30) = 28591.44/\lambda_{max} \quad (1)$$

$$\pi* = 0.314\ \left(27.52 - \upsilon_{N,N-dimethyl\ -4\ -nitroaniline}\right) \quad (2)$$

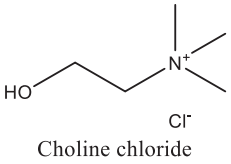
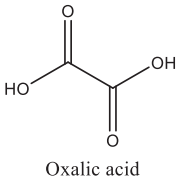
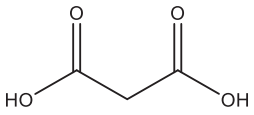
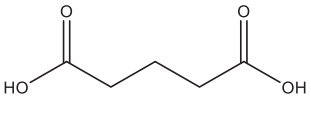
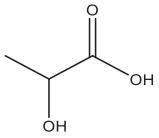
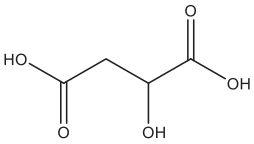
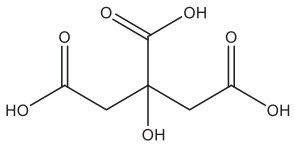
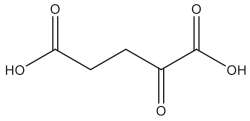
$$\beta = \frac{\left(1.305\upsilon_{N,N-dimethyl\ -4\ -nitroaniline} + 2.64 - \upsilon_{4\ -nitroaniline}\right)}{2.8} \quad (3)$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \upsilon_{NR}}{1.6078} \quad (4)$$

$$\upsilon = 1/\left(\lambda_{max} \times 10^{-4}\right) \quad (5)$$

Where λ_{max} (nm) represents the maximum absorption wavelength and E_T (30) refers to the transition energy, respectively.

Table 1
Molecular structures of the HBA and HBD investigated in this work.

HBA	HBD
 Choline chloride	 Oxalic acid
	 Malonic acid
	 Glutaric acid
	 Lactic acid
	 Malic acid
	 Citric acid
	 α-Ketoglutaric acid

2.2.5. Computational Details

Based on our previous theoretical study of DESs [16], the ChCl compounds, HBD molecules, and ChCl plus HBD complexes (DES minimal clusters) were optimized with the DFT method. In order to obtain the optimized geometries at the minimum of energy, molecular docking simulation was conducted through the YASARA v17.4.17 program [15]. The remaining unique stable geometries were subjected to re-optimization at the M06-2X density functional, combined with the 6-31++ g (d, p) basis set [20]. Recent studies have demonstrated the superior performance of M06-2X density functional in accurately describing hydrogen bond interactions [21,22]. To eliminate the dispersion effect, we added the keyword of empirical dispersion = D3 in the route section, and all calculations were optimized under no symmetry restrictions. The H-bonds distances and angles of DESs have been calculated and used to quantify H-bonded interactions. Furthermore, based on electron density and its derivatives, atoms in molecules (AIM) theory and reduced density gradient (RDG) analysis were applied to confirm the presence of the noncovalent interactions, such as hydrogen bonds and van der Waals interactions [23]. Simultaneously, the electrostatic potential analysis (ESP) determined the spatial distribution of electrostatic potential, which provides information about the electrostatic interactions in the system [24]. Various software packages were used for the evaluation of the specific properties and visualization, namely, Multiwfn 3.8 and VMD was used [25].

3. Results and Discussion

3.1. Structures of DESs

Seven carboxylic acids with different carbon chain lengths and functional groups were selected to describe the relationship between the structure and properties of DES formed between ChCl and carboxylic acids. Since our study involves an extensive hydrogen bonding, it is very important to explore the NMR spectrum to verify their chemical structure and understand the mechanism of the interactions among the components of DESs [26]. As shown in **Table S1**, the ^{1}H NMR spectrum of ChCl shows the characteristic methyl group signals at 3.13 ppm, methylene protons at 3.45 ppm and 3.76 ppm, and hydroxyl proton at 5.72 ppm. The chemical shift of N-methylene and methyl groups of ChCl in the ^{1}H NMR spectrum of DESs gets more shielded towards upfield due to the induced electron density by carboxylic acids (**Figure S1-S8**). Similarly, the hydroxyl proton of the ChCl-acid spectrum shows an appreciable chemical shift in presence of carboxylic acids due to the electronic transfer via the hydrogen bonding network. Simultaneously, the ^{13}C NMR spectrum of ChCl shows three signals at 53.52, 55.45, and 67.33 ppm assigned to the methyl, methylene, and carbon linked to the hydroxyl group, respectively (**Table S2**). Comparing with the ^{13}C NMR spectrum of ChCl, it could be found that the three methyl carbons in carboxylic acid-based DESs generate downfield chemical shifts with $\Delta\delta$ between 0.3 and 0.4 ppm, respectively (**Figure S9-S16**). This phenomenon can be interpreted that the density of electron clouds around the N atom increases, which in turn leads to the reduction of electron clouds density around the three methyl carbons, thereby enhancing the ^{13}C NMR signal [27]. Therefore, the above results confirm the formation of hydrogen bonds between ChCl and carboxylic acid.

In order to check the interaction between the proton in the choline part and the carboxylic carbon of the acid part, DESs were further characterized by ^{1}H–^{1}H NOESY techniques [28]. As shown in Fig. 1**A**, taking ChCl-Glu as an example, the NOESY spectrum shows that there is a strong interaction between the carboxyl proton in the Glu molecule with the hydroxyl group and methylene protons in ChCl, which means that the carboxyl group of Glu could form a hydrogen bond force with the choline. The above NMR results illustrate that molecular interaction exists between the HBA and HBD, and the hydroxyl hydrogen atom in ChCl interacts with carboxyl groups in organic acids to form hydrogen bonds.

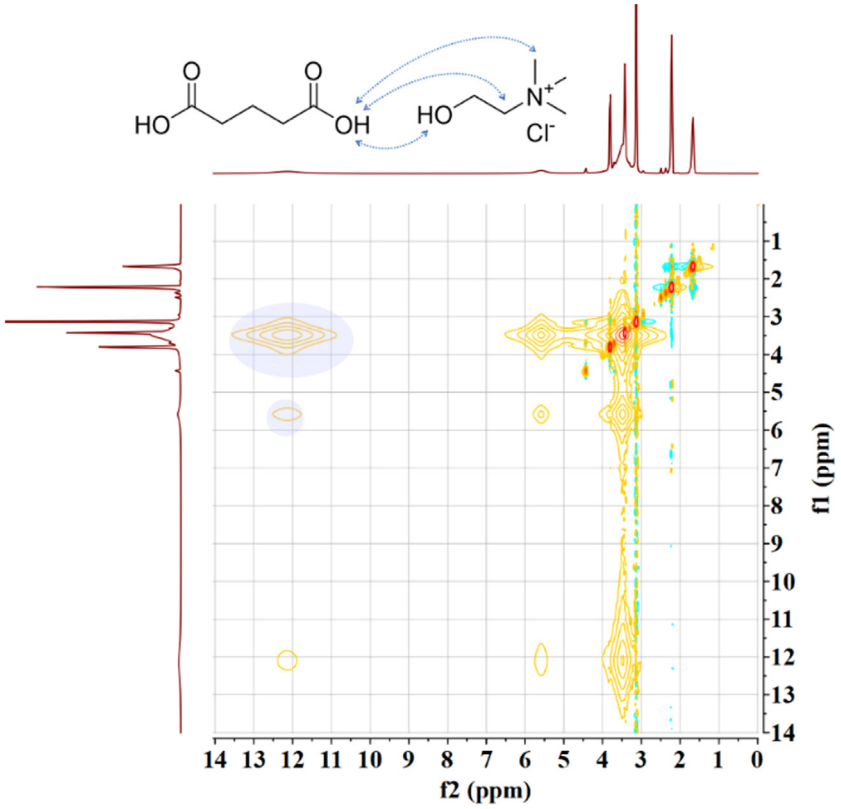


Fig. 1. ^{1}H–^{1}H NOESY spectra of ChCl-Glu (1:1) DES.

3.2. Physicochemical properties

3.2.1. Viscosity and electrical conductivity

The viscosity describes the internal resistance of a fluid to shear stress and affects the mass and heat transfer of the reactions [29]. In addition, electrical conductivity indicates the movement of electric charges and is important for their applications in electrochemistry [30]. The viscosity results of ChCl-carboxylic acids DESs obtained in this work were presented in Figs. 2**A and** 2**B**. The wide range of viscosity is usually attributed to the formation of hydrogen bond interactions between ChCl and carboxylic acid [9]. The highest viscosity belongs to the ChCl-Cit DES mixture, which might be due to the contribution of three carboxyl and one hydroxyl group in the citric acid molecule. This leads to the strongest hydrogen bonding in ChCl-Cit system, which is confirmed in the calculation part. It is worthy to mention that viscosities of DESs synthesized from hydroxyl acids as the HBDs are generally lower than those of DESs with an extra carboxylic group or extra hydroxyl group. The viscosities of ChCl-Oxa /Malo /Glu (1:1) are 3401, 563, and 790 mPa·s at 313.15 K, respectively. Regarding the size of the alkyl chain, usually, an increase in the size of the hydrocarbon chain leads to an increase in viscosity. However, ChCl-Oxa with the shorter alkyl chain is more viscous in the results. Hunt et al. reported a similar phenomenon and attributed this type of behavior to the smaller molecular weight of Oxa [31].

Figs. 2**C and** 2**D** show the electrical conductivity of carboxylic acid-based DESs. It follows the reverse trend to viscosity as may be expected. Owing to the relatively high viscosities, most ChCl-acid DESs show relatively poor conductivity. For the studied DESs, the electrical conductivity decreases in the order ChCl-La > ChCl-Malo > ChCl-Glu > ChCl-Oxa > ChCl-Ket > ChCl-Mali > ChCl-Cit at all temperatures, in agreement with the observed increasing trend of viscosity. Besides, it can be observed that the conductivity of the DESs depends on the structure of carboxylic acid. Higher conductivity values are obtained for ChCl-La, ChCl-Malo, and ChCl-Glu, which are ascribed to the relatively weak hydrogen bonds in these DESs. In the case of DESs formed by carboxylic acids, the electrical conductivity values decrease with an increasing number of hydroxyl groups and carboxylic groups.

The temperature has a major effect on the viscosity and electrical conductivity of ChCl-Acid mixtures (Fig. 2). As the temperature increases, the viscosity decreases and the electrical conductivity increases rapidly. Since the interactions among molecules are weakened with the increase of temperature, the internal resistance of the molecules decreases, resulting in the increase of the electrical conductivity of the system. The variation in viscosity with temperature is a nonlinear behavior and generally follows the Arrhenius equation:

$$\ln\ \eta = \ln\ \eta_{\infty} + \frac{E_{\eta}}{RT} \tag{6}$$

Similar to viscosity, conductivity data have been fitted to an Arrhenius equation:

$$\ln\ \sigma = \ln\ \sigma_{\infty} - \frac{E_{\sigma}}{RT} \tag{7}$$

Where η represents the experimental viscosity, mPa·s. R is the gas constant, 8.314 J·mol^{-1}·K^{-1}. And T is the temperature, K. σ represents the electrical conductivity, mS·cm^{-1}. η_{∞} (mPa·s) and σ_{∞} (mS·cm^{-1}) are the viscosity and conductivity at infinite temperature, respectively. E_{η} and E_{σ} (J/mol) are the activation energies, J/mol.

The parameters ln η_{∞} and ln σ_{∞} depend on the structure of the HBA and HBD. The activation energy describes the potential barrier that ions must have to overcome to move past each other. The higher activation energy, the more difficult for the ions to move past each other [32]. In this work, the present measured E_{η}

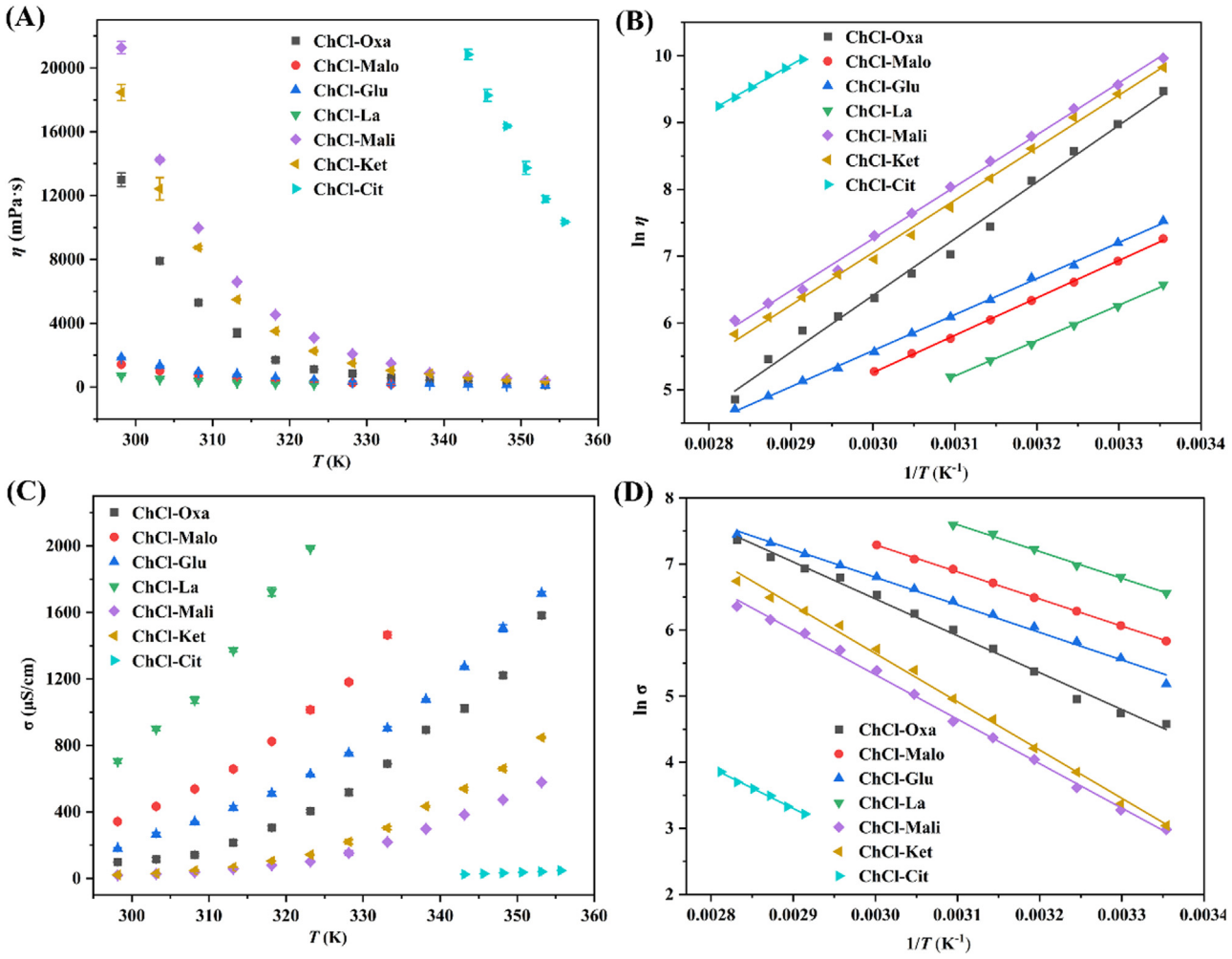


Fig. 2. Variations of viscosity and conductivity with temperature.

Table 2
Correlation parameters by Eq. (6) and Eq. (7) for the viscosities and conductivities of ChCl-Acid DESs as a function of temperature.

Sample	ln η_∞	E_η (kcal/mol)	R_η^2	ln σ_∞	E_σ (kcal/mol)	R_σ^2
ChCl-Oxa	-19.062	16.873	0.9918	23.229	11.099	0.9931
ChCl-Malo	-11.541	11.125	0.9993	19.602	8.154	0.9993
ChCl-Glu	-10.608	10.726	0.9989	19.313	8.289	0.9957
ChCl-La	-11.235	10.538	0.9987	20.169	8.061	0.9971
ChCl-Mali	-16.603	15.449	0.9973	25.516	13.375	0.9973
ChCl-Ket	-16.514	15.609	0.9970	27.568	14.521	0.9958
ChCl-Cit	-10.310	13.822	0.9948	21.170	12.245	0.9942

and E_σ values are shown in Table 2. Different E_η and E_σ values are observed for the studied DESs, probably due to the nature of each HBD constituent. For the seven studied carboxylic-based DESs, the values of the E_η range from 10.538 to 16.873 kcal/mol, among which ChCl-Oxa shows the highest E_η value (16.873 kcal/mol). The values of E_σ range from 8.061 to 14.521 kcal/mol. It is interesting that the largest ln η_∞ value is obtained for ChCl-Cit, which is consistent with its viscosity data. Therefore, intermolecular interactions play a pivotal role in DES viscosity and electrical conductivity.

Table 3
Solvatochromic parameters of the DES mixtures.

Sample	E_T (30)	E_T^N	π^*	α	β
ChCl-Oxa	47.660	0.523	1.157	1.313	0.543
ChCl-Malo	47.907	0.531	1.115	1.286	0.550
ChCl-Glu	48.458	0.548	1.063	1.199	0.568
ChCl-La	48.723	0.556	1.115	1.109	0.541
ChCl-Mali	47.795	0.528	1.168	1.276	0.439
ChCl-Ket	48.020	0.535	1.154	1.237	0.413

3.2.2. Solvatochromic parameters

Another important physicochemical property is the Kamlet-Taft parameter. It is a key factor to reflect their overall solvation capability towards a solvent which affects reaction rate, product yield, and selectivity as well as reaction mechanism during a chemical reaction process. Because of the high viscosity of the ChCl-Cit mixture, it is not possible to measure the solvatochromic parameters at room temperature (25°C). Table 3 shows the solvatochromic pa-

rameters of the other six ChCl-acid DESs mixtures studied in this work.

Nile red is selected as a solvatochromic probe in acidic DESs due to its low alkalinity (pKa = 1.00 ± 0.05) and photochemical stability [14]. The solvatochromic parameter E_T^N is a normalized scale of E_T (30). From the results, the λ_{max} of Nile red does not shift significantly among the various DESs. The E_T^N values of the DESs mixtures studied in this work are all between 0.5 and 0.6, which are slightly lower than those of short-chain alcohols (e.g., the value for methanol is 0.762) [33]. Increasing the carbon chain of the HBD increases the value of the E_T^N parameter and thus reduces the polarity of the DESs.

The parameter π^* describes the combination of polarity and polarizability [14]. The π^* values of the DESs mixtures increase in the following order: ChCl-Oxa < ChCl-Malo < ChCl-Glu. The π^* values decrease in increasing the alkyl chain length of HBD due to the presence of permanent dipoles and delocalized bonds in the solvent [14], which is consistent with the reports by Teles et al [34]. It is clear that the dominant species in DESs ruling the values of π^* is the HBD. The ChCl-Mali shows a higher π^* value than ChCl-Cit, indicating that too many carboxyl groups decrease the mixture polarizability.

For the solvent, the parameter β is a measure of the hydrogen-bond basicity and represents an ability to donate electron density to form a hydrogen bond with a solute's protons [35]. The β values of tested DESs vary between 0.4 and 0.6, with the lowest value of ChCl-Ket (0.413) and the highest value of ChCl-Glu (0.568). By comparison, the effects of alkyl chain length and functional group of HBDs on β values of DESs are opposite to that on their π^* values.

The α scale is a measure of solvent hydrogen bond acidity. From the results, the α values of DESs obtained here range from 1.1 to 1.4, indicating that those DESs are hydrogen bond acidic solvents [35]. The alkyl chain of the HBD has a significant impact on the α parameter, and its increasing order is: ChCl-La< ChCl-Glu < ChCl-Ket < ChCl-Mali< ChCl-Malo< ChCl-Oxa. Obviously, the effect of HBD carbon chain length on α values of DESs has the same trend as that on the π^* values of DESs. Due to the presence of carboxylic functional group, all the tested DESs possess relatively high α values, indicating that these DESs have high hydrogen bond donating ability.

3.3. Computational Results

3.3.1. Geometries analysis

The docking results of ChCl and carboxylic acids show a variety of possible conformations after the cluster analysis of YASARA. All of the representative conformers are plotted in **Table S3**. According to the scoring order, the energy-minimized structure was taken as the optimal one and then re-optimization with Gaussian at the M06-2X/6-31++ g (d, p) level. The hydrogen bond distances and bond angles of DESs were obtained after frequency correction (**Figure S17**). The primary interactions in these DESs are those between the chloride anion and the acid OH group ($Cl^-\cdots H-O_{Acid}$), choline and chloride charge centres ($Cl^-\cdots H-O_{Ch}{}^+$), the choline and acid OH groups ($O_{Ch}{}^+\cdots H-O_{Acid}$), and the acid OH group self-interactions ($O_{Acid}\cdots H-O_{Acid}$). Among those interactions, the strongest is $Cl^-\cdots H-O_{Acid}$ hydrogen bond. **Figure S17** demonstrates that chloride ions are always close to the carboxyl part of organic acids, indicating that the interaction between chloride ions and carboxyl groups is stronger compared with hydroxyl groups. Moreover, relative to the calculated geometry for pure choline chloride, the distance between the chloride anion and the choline headgroup increases to a large extent. To investigate the effect of HBD structure, we compared the bond angles of Cl^- and hydroxyl atoms in organic acids in the optimized DESs. The lowering of those bond angles (∠ClHO) shows that molecular interaction occurs between the HBA and the HBD during DES formulation. As evident from Table 4, the ∠ClHO angles of most DESs are close to 180°, indicating linear hydrogen bonds [36]. The ∠ClHO angles of ChCl-Mali and ChCl-Cit deviate to some extent from linearity, with 167° and 141°, respectively. This may be due to the presence of an additional O−H···O hydrogen bonding interaction between the carbonyl and hydroxyl oxygen atoms in the HBD.

Also, the strength of hydrogen bonding can be quantified by measuring the bond distance. As can be judged from the hydrogen bond distances, chloride ion prefers to be positioned near H atoms associated with the hydroxyl group of the organic acid. Stronger interactions are present between the carboxylic proton of the HBD and the chloride anion of the HBA ($Cl^-\cdots H-O$Acid), which can be attributed to the larger electronegativity of the chloride ion. These hydrogen-bond distances in these DESs are within 1.895 ≤ Å ≤ 2.19, which can form a short-strong hydrogen bonding [14]. Interestingly, the chloride ion can form hydrogen bonds with the two carboxyl groups of citric acid in ChCl-Cit DES and the bond angles are both 141°, which is the leading cause of the increase in hydrogen bond interactions. Meanwhile, an important point to note is that the bond length increases as the chain length in carboxylic acid increases.

3.3.2. The Laplacian of the Electron Density of the Bond Critical Point (BCPs)

In atoms in molecules (AIM) theory, the topological parameters at the bonding critical point (BCP), such as electron density (ρ), Laplace function of electron density ($\nabla^2\rho$), potential density (V_{cp}), and the energy of hydrogen bond (E_{HB}) can be used to analyze the nature of hydrogen bonds and estimate their strength [27]. The topological analyses of the electron densities show the presence of a BCP between the chloride ion and hydroxyl atoms in organic acids in the DESs (**Figure S18 and** Table 5). Besides, when $\nabla^2\rho$ is greater than 0, it represents the presence of hydrogen bonds or ionic bonds, or van der Waals forces. All our proposed hydrogen bonds have a positive $\nabla^2\rho$ and their values lie between 0.0618 and 0.0729, demonstrating the existence of H-bonds in DESs.

Recently, Emamian et al. have developed new equations for quickly and reliably predicting hydrogen bond energy for charged H-bond complexes [37]:

$$E_{HB}(\text{kcal/mol}) = -332.34 \times \rho_{BCP}/\text{a.u.} - 1.0661 \quad (8)$$

Small geometrical changes of HBD lead to little energy change of E_{HB} (-15.001 kcal/mol to -18.161 kcal/mol). In our case, ChCl-Cit DES has the highest E_{HB}, that is, it has the strongest interaction energy, which is consistent with our previous study [16]. ChCl-Glu DES has the lowest E_{HB} value in comparison to ChCl-Oxa and ChCl-Malo, thereby indicating that the bond energy of $Cl^-\cdots H-O$Acid decreases with the increase of carbon chain length and the former exhibits the highest hydrogen bond donating ability. For dicarboxylic acids, the E_{HB} value of hydroxyl acid based DESs is lower than those of DESs with an extra carboxylic or hydroxyl group.

In order to shed light on the interaction between the HBA and HBD, we carried out RDG analysis to further discuss noncovalent interactions. Based on the electron density and its derivatives, RDG isosurfaces with different colors reveal the underlying rich representation of hydrogen bonds, van der Waals interactions, and steric repulsion in molecular complexes [38]. The strength and the type of interactions of the HBA/HBD can be visualized with the RDG plots (Fig. 3). The mapped color indicates the type of interaction: the blue region shows strong interaction force, the green region shows weak interactions like van der Waals forces, and the steric effect is distinguished by red. From the RDG isosurface map, the blue region between the carbonyl ion and hydroxyl oxygen atoms in HBD is also vividly exhibited. It seems that H-bonds domi-

Table 4
The hydrogen bond distances (R, Å), and bond angles (deg) for the studied DESs.

Sample	R (Cl···H−O_{Acid})	R (Cl···H−O_{Ch}^+)	R (O_{Ch}^+···H−O_{Acid})	∠ClHO (°)
ChCl-Oxa	1.898	2.309	/	178
ChCl-Malo	1.925	2.310	/	176
ChCl-Glu	1.961	2.295	/	175
ChCl-La	1.924	2.303	/	175
ChCl-Mali	1.895	2.365	/	167
ChCl-Ket	1.897	2.309	/	177
ChCl-Cit	2.216, 2.195	/	1.87414	141

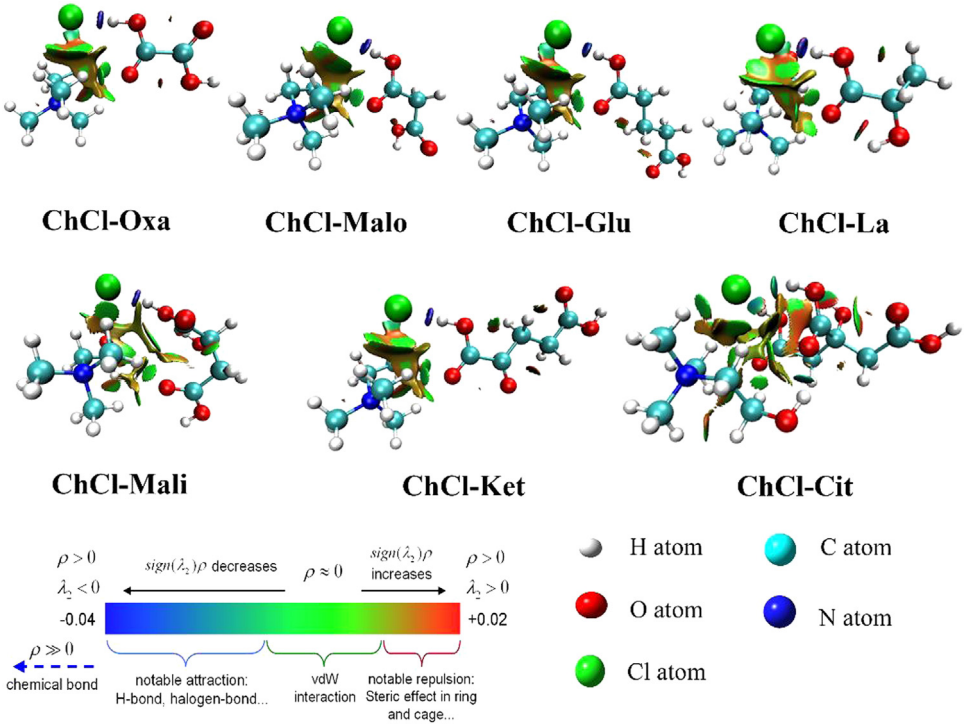


Fig. 3. Isosurface map of RDG of seven DESs. The surfaces are colored on a red-green-blue scale according to values of sign(λ_2) ρ, ranging from −0.04 to +0.02 a.u.

Table 5
Topological parameters at bond critical points (BCPs) for hydrogen bonds (Cl···H−OAcid).

Sample	$\rho_{BCP(a.u.)}$	$\nabla^2\rho_{BCP(a.u.)}$	V (a.u.)	E_{HB}(kcal/mol)
ChCl-Oxa	0.0506	0.0618	-0.0435	-17.432
ChCl-Malo	0.0474	0.0652	-0.0399	-16.381
ChCl-Glu	0.0431	0.0690	-0.0352	-15.001
ChCl-La	0.0474	0.0655	-0.0400	-16.387
ChCl-Mali	0.0509	0.0669	-0.0394	-17.521
ChCl-Ket	0.0509	0.0729	-0.0336	-17.510
ChCl-Cit	0.0529	0.0713	-0.0344	-18.161

nate the interaction between the HBA and HBD. All RDG isosurface maps in the region between carboxylic acid and chloride ion are large green isosurface, confirming that van der Waals forces also represent the main contribution to the weak interactions in these DESs systems. Meanwhile, the RDG analysis also exhibits the unnegligible repulsion between excess hydroxyl and carboxyl groups in acid. Among the studied DESs, ChCl-Cit DESs show stronger hydrogen bonds, van der Waals force, and steric hindrance effects, which are attributed to the more carboxyl and hydroxyl structures in Cit.

3.3.3. ESP analysis

ESP analysis on the molecular surface has been prevalently used to predict intermolecular interaction and visualize the total charge distribution and relative polarity of the studied DESs structures [39]. The site possessing more negative ESP has the stronger ability to attract electrophiles and thus is more possible to be the reactive site. As depicted in Fig. 4, the electronegative area is around the Cl^- and O atoms in ChCl molecule, whereas the electropositive area is found around H atoms. For La, the electronegative area is also around O atoms, whereas the electropositive area is around the H atom. For ChCl-La, it has bluer isosurfaces surrounding the molecule. This corresponds to stronger bonds between multiple carboxyl and hydroxyl groups. Furthermore, the negative potential of the carboxylic acid is easily transferred from the Cl^-···H−OAcid hydrogen bond to the additional positive potential around the ChCl. The results show that the electronegative area of ChCl will be close to the electropositive area of acid during the formation of the monomer model of DES, and vice versa. The electrostatic interaction may be mainly present at the Cl^-···H−OAcid, which is consistent with our previous bond length analysis. Several ESP extreme points are distributed around the carboxyl group, reflecting the polar portion of the HBD (**Figure S19**). As observed, the negative charge of the O atom in the organic acid stabilizes the positive charge at the corresponding position of the DES structure in most cases. This observation well explains why the carboxyl groups of the HBD structure form H-bonds with ChCl prior to the other groups in adjacent monomers [40].

Previously, Lu et al. defined a descriptor named molecular polarity index (MPI), which is expressed as the following equation

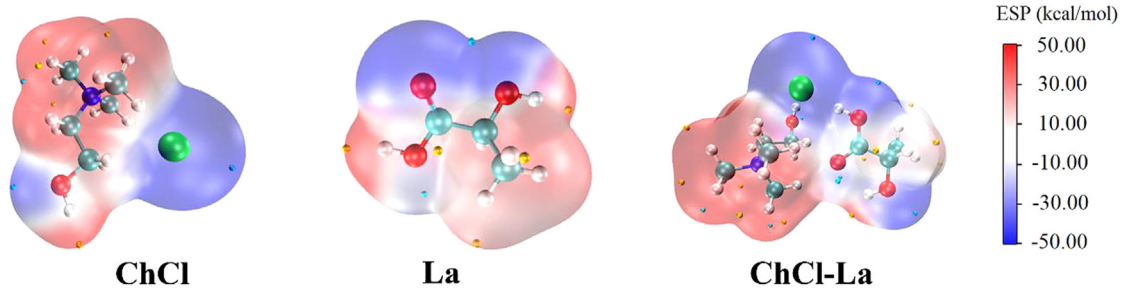


Fig. 4. ESP surface map visualization for ChCl, La, and ChCl-La structures. ESP ranges are included in the legend at the figure panel.

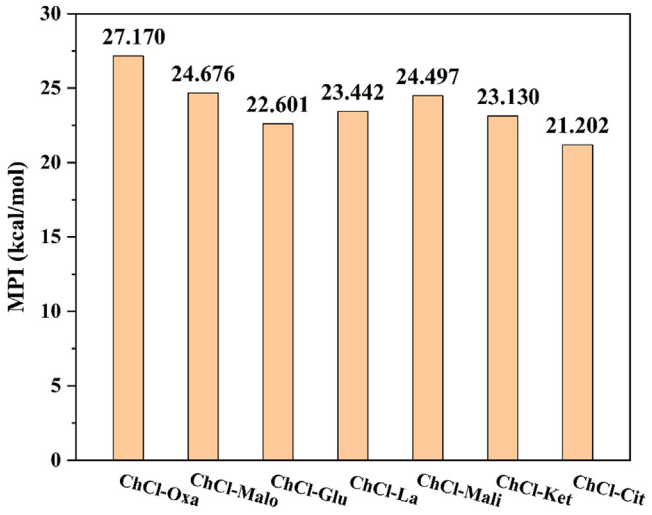


Fig. 5. The MPI values for studied DESs structures.

[41]:

$$\mathrm{MPI} = (1/\mathrm{A}) \iint \mathrm{SV(r)}|D\mathrm{s} \tag{9}$$

where A and $V(\mathrm{r})$ refer to the area of van der Waals surface and value of ESP at a point r in space, respectively. The larger the MPI, the greater the overall polarity of the molecule. Because the uneven charge distribution of the system is the reflection of molecular polarity, the more uneven the distribution, the more positive or negative the electrostatic potential of the molecular surface will appear, resulting in a larger MPI. As observed in Fig. 5, the surface region with the largest MPI value corresponds to ChCl-Oxa DES, which is consistent with the lowest E_T^{N} value determined in our experiment. The maximum polarity of ChCl-Oxa DES is due to its small molecular weight. Similarly, the MIP index decreases with increasing carbon chain length. The lowest MPI value is ChCl-Cit, which is affected by the more carboxyl and hydroxyl groups in the Cit structure. This is because the lone-pair electrons of the O atom contribute significantly to the negative value of the ESP, which causes the shared electrons in the covalent bond to shift toward the O atom and negative charges to accumulate around it. The theoretical ESP and MPI index calculations verify the experimental results, further implying that the strength of the electrostatic interaction between the carboxylic acid proton and ChCl should be fairly strong.

4. Conclusions

An experimental and theoretical study were performed on seven ChCl-carboxylic acid DESs for their structure, properties, and interactions between the components of DESs. The intermolecular interaction was confirmed by ^{1}H NMR, ^{13}C NMR, and 2D NMR spectrum analysis, mainly due to hydrogen bonds between the carboxyl groups of organic acids and the choline hydroxyl groups. In agreement with the NMR experimental evidence, the DFT theoretical studies also verified this conclusion from the viewpoint of molecular structure. This work shows that the functional groups of HBD, the number of hydroxyl and carboxyl groups, as well as the length of the carbon chain, have a considerable influence on the physicochemical properties of DESs. Among the seven studied carboxylic-based DESs, ChCl-Cit DES has the highest viscosity and lowest conductivity due to a higher number of functional groups than the other acids and shows higher internal interaction energy. Besides, the solvation parameters indicate that the polarity of DESs decreases with increasing the alkyl chain length of HBD, while the value of hydrogen bond acidity is the opposite, which is consistent with the theoretical MPI results. Furthermore, RDG and ESP analysis show that the hydrogen bonding and van der Waals forces play an important role in the formation of DESs. Based on the structure of different carboxylic acids, our work systematically analyzed the relationship between the structure and properties of DESs. Comparative analysis of experimental and theoretical results gives us a deep understanding of structure–properties relationships as well as further provide more references for the designing of DESs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Man Zhang: Writing – original draft, Formal analysis. **Xiaomeng Zhao:** Data curation, Writing – original draft. **Siyang Tang:** Software, Validation. **Kejing Wu:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Binshen Wang:** Methodology, Validation, Data curation, Visualization. **Yingying Liu:** Writing – review & editing. **Yingming Zhu:** Visualization, Investigation. **Houfang Lu:** Writing – review & editing. **Bin Liang:** Supervision.

Data availability

The data that has been used is confidential.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 21878190) and the Cooperation Fund of Sichuan University and The People's Government of Yibin City (No. 2020CDYB-34). The authors would like to thank the Center of Engineering Experimental Teaching, School of Chemical Engineering, Sichuan University for providing Nuclear Magnetic Resonance.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.molstruc.2022.134283.

References

[1] M. Francisco, A. van den Bruinhorst, M.C. Kroon, Low-transition-temperature mixtures (LTTMs): A new generation of designer solvents, Angew Chem Int Ed Engl 52 (2013) 3074–3085, doi:10.1002/anie.201207548.

[2] M.A.R. Martins, S.P. Pinho, J.A.P. Coutinho, Insights into the nature of eutectic and deep eutectic mixtures, J. Solution Chem. 48 (2019) 962–982, doi:10.1007/s10953-018-0793-1.

[3] R. Craveiro, I. Aroso, V. Flammia, T. Carvalho, M.T. Viciosa, M. Dionísio, S. Barreiros, R.L. Reis, A.R.C. Duarte, A. Paiva, Properties and thermal behavior of natural deep eutectic solvents, J. Mol. Liq. 215 (2016) 534–540, doi:10.1016/j.molliq.2016.01.038.

[4] Q. Zhang, K. De Oliveira Vigier, S. Royer, F. Jerome, Deep eutectic solvents: syntheses, properties and applications, Chem Soc Rev 41 (2012) 7108–7146, doi:10.1039/c2cs35178a.

[5] L. Bahadori, M.H. Chakrabarti, F.S. Mjalli, I.M. Alnashef, N.S.A. Manan, M.A. Hashim, Physicochemical properties of ammonium-based deep eutectic solvents and their electrochemical evaluation using organometallic reference redox systems, Electrochim. Acta. 113 (2013) 205–211, doi:10.1016/j.electacta.2013.09.102.

[6] F. Bergua, I. Delso, J. Muñoz-Embid, C. Lafuente, M. Artal, Structure and properties of two glucose-based deep eutectic systems, Food Chem 336 (2021) 127717, doi:10.1016/j.foodchem.2020.127717.

[7] Y. Dai, G.J. Witkamp, R. Verpoorte, Y.H. Choi, Natural deep eutectic solvents as a new extraction media for phenolic metabolites in carthamus tinctorius L, Anal. Chem 85 (2013) 6272–6278, doi:10.1021/ac400432p.

[8] C. Alvarez-Vasco, R. Ma, M. Quintero, M. Guo, S. Geleynse, K.K. Ramasamy, M. Wolcott, X. Zhang, Unique low-molecular-weight lignin with high purity extracted from wood by deep eutectic solvents (DES): A source of lignin for valorization, Green Chem 18 (2016) 5133–5141, doi:10.1039/c6gc01007e.

[9] M. Zhang, X.Y.L. Zhang, Y.Y. Liu, K.J. Wu, Y.M. Zhu, H.F. Lu, B. Liang, Insights into the relationships between physicochemical properties, solvent performance, and applications of deep eutectic solvents, Environ. Sci. Pollut. Res. 28 (2021) 35537–35563, doi:10.1007/s11356-021-14485-2.

[10] H.P. Li, Y.H. Chang, W.S. Zhu, C.W. Wang, C. Wang, S. Yin, M. Zhang, H.M. Li, Theoretical evidence of charge transfer interaction between SO_2 and deep eutectic solvents formed by choline chloride and glycerol, Phys. Chem. Chem. Phys. 17 (2015) 28729–28742, doi:10.1039/c5cp04172d.

[11] W. Jiang, K. Zhu, H.P. Li, L.H. Zhu, M.Q. Hua, J. Xiao, C. Wang, Z.Z. Yang, G. Chen, W. Zhu, H.M. Li, S. Dai, Synergistic effect of dual Brønsted acidic deep eutectic solvents for oxidative desulfurization of diesel fuel, Chem. Eng. J. 394 (2020) 124831, doi:10.1016/j.cej.2020.124831.

[12] G. García, S. Aparicio, R. Ullah, M. Atilhan, Deep eutectic solvents: Physicochemical properties and gas separation applications, Energy & Fuels 29 (2015) 2616–2644, doi:10.1021/ef5028873.

[13] R. Gautam, N. Kumar, J.G. Lynam, Theoretical and experimental study of choline chloride-carboxylic acid deep eutectic solvents and their hydrogen bonds, J. Mol. Struct. 1222 (2020) 128849, doi:10.1016/j.molstruc.2020.128849.

[14] A.K. Dwamena, D.E. Raynie, Solvatochromic parameters of deep eutectic solvents: effect of different carboxylic acids as hydrogen bond donor, J. Chem. Eng. Data. 65 (2020) 640–646, doi:10.1021/acs.jced.9b00872.

[15] M. Saha, M. Saha, M.S. Rahman, M.N. Hossain, D.E. Raynie, M.A. Halim, Molecular and spectroscopic insights of a choline chloride based therapeutic deep eutectic solvent, J. Phys. Chem. A. 124 (2020) 4690–4699, doi:10.1021/acs.jpca.0c00851.

[16] M. Zhang, R.B. Tian, H. Han, K.J. Wu, B.S. Wang, Y.Y. Liu, Y.M. Zhu, H.F. Lu, B. Liang, Preparation strategy and stability of deep eutectic solvents: A case study based on choline chloride-carboxylic acid, J. Clean. Prod. 345 (2022) 131028, doi:10.1016/j.jclepro.2022.131028.

[17] C. Laurence, P. Nicolet, M.T. Dalati, J.L.M. Abboud, R. Notario, The empirical treatment of solvent-solute interactions: 15 Years of π, J. Phys. Chem. 98 (1994) 5807–5816, doi:10.1021/j100074a003.

[18] Q.L. Liu, X.H. Zhao, D.K. Yu, H.T. Yu, Y.B. Zhang, Z.M. Xue, T.C. Mu, Novel deep eutectic solvents with different functional groups towards highly efficient dissolution of lignin, Green Chem 21 (2019) 5291–5297, doi:10.1039/c9gc02306b.

[19] W. Yusong, T. Sasaki, K. Kazushi, T. Seo, K. Sakurai, Interactions between spiropyrans and room-temperature ionic liquids: Photochromism and solvatochromism, J. Phys. Chem. B. 112 (2008) 7530–7536, doi:10.1021/jp800957c.

[20] Y. Zhao, D.G. Truhlar, Density functionals with broad applicability in chemistry, Acc. Chem. Res. 41 (2008) 157–167, doi:10.1021/ar700111a.

[21] P.L. Pisano, M. Espino, M. de los Á. Fernández, M.F. Silva, A.C. Olivieri, Structural analysis of natural deep eutectic solvents. Theoretical and experimental study, Microchem. J. 143 (2018) 252–258, doi:10.1016/j.microc.2018.08.016.

[22] S. Kozuch, J.M.L. Martin, Halogen bonds: Benchmarks and theoretical analysis, J. Chem. Theory Comput. 9 (2013) 1918–1931, doi:10.1021/ct301064t.

[23] T. Lu, Q. Chen, Interaction region indicator: A simple real space function clearly revealing both chemical bonds and weak interactions**, Chemistry Methods 1 (2021) 231–239, doi:10.1002/cmtd.202100007.

[24] T. Lu, F.W. Chen, Quantitative analysis of molecular surface based on improved marching tetrahedra algorithm, J. Mol. Graph. Model. 38 (2012) 314–323, doi:10.1016/j.jmgm.2012.07.004.

[25] T. Lu, F.W. Chen, Multiwfn: a multifunctional wavefunction analyzer, J. Comput. Chem. 33 (2012) 580–592, doi:10.1002/jcc.22885.

[26] A.K. Jangir, H. Mandviwala, P. Patel, S. Sharma, K. Kuperkar, Acumen into the effect of alcohols on choline chloride: L-lactic acid-based natural deep eutectic solvent (NADES): A spectral investigation unified with theoretical and thermophysical characterization, J. Mol. Liq. 317 (2020) 113923, doi:10.1016/j.molliq.2020.113923.

[27] H. Fu, X.Y. Wang, H.N. Sang, Y.P. Hou, X.H. Chen, X. Feng, Dissolution behavior of microcrystalline cellulose in DBU-based deep eutectic solvents: insights from spectroscopic investigation and quantum chemical calculations, J. Mol. Liq. 299 (2020) 112140, doi:10.1016/j.molliq.2019.112140.

[28] Y. Dai, J. van Spronsen, G.J. Witkamp, R. Verpoorte, Y.H. Choi, Natural deep eutectic solvents as new potential media for green technology, Anal. Chim. Acta 766 (2013) 61–68, doi:10.1016/j.aca.2012.12.019.

[29] Y.Y. Liu, S. Lin, H.F. Lu, C.J. Liu, K.J. Wu, W. Jiang, B. Liang, Studies on viscosity and conductivity of 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU)-glycerol and CO_2-DBU-glycerol solutions at temperatures from 288.1 K to 328.1 K, J. Chem. Thermodyn. 136 (2019) 16–27, doi:10.1016/j.jct.2019.04.017.

[30] D.Y. Yue, Y. Jing, J. Ma, Y. Yao, Y.Z. Jia, Physicochemical properties of ionic liquid analogue containing magnesium chloride as temperature and composition dependence, J. Therm. Anal. Calorim. 110 (2012) 773–780, doi:10.1007/s10973-011-1960-4.

[31] X.Y. Li, M.Q. Hou, B.X. Han, X.L. Wang, L.Z. Zou, Solubility of CO_2 in a choline chloride + urea eutectic mixture, J. Chem. Eng. Data. 53 (2008) 548–550, doi:10.1021/je700638u.

[32] F. Chemat, H. Anjum, A.M. Shariff, P. Kumar, T. Murugesan, Thermal and physical properties of (Choline chloride + urea + l-arginine) deep eutectic solvents, J. Mol. Liq. 218 (2016) 301–308, doi:10.1016/j.molliq.2016.02.062.

[33] A. Pandey, R. Rai, M. Pal, S. Pandey, How polar are choline chloride-based deep eutectic solvents? Phys. Chem. Chem. Phys. 16 (2014) 1559–1568, doi:10.1039/c3cp53456a.

[34] A.R.R. Teles, E.V. Capela, R.S. Carmo, J.A.P. Coutinho, A.J.D. Silvestre, M.G. Freire, Solvatochromic parameters of deep eutectic solvents formed by ammonium-based salts and carboxylic acids, Fluid Phase Equilib 448 (2017) 15–21, doi:10.1016/j.fluid.2017.04.020.

[35] X.Q. Liang, Y. Zhu, B.K. Qi, S.Q. Li, J.Q. Luo, Y.H. Wan, Structure-property-performance relationships of lactic acid-based deep eutectic solvents with different hydrogen bond acceptors for corn stover pretreatment, Bioresour. Technol. 336 (2021) 125312, doi:10.1016/j.biortech.2021.125312.

[36] L.E. Shmukler, I.V. Fedorova, M.S. Gruzdev, L.P. Safonova, Triethylamine-based salts: protic ionic liquids or molecular complexes? J. Phys. Chem. B. 123 (2019) 10794–10806, doi:10.1021/acs.jpcb.9b08032.

[37] S. Emamian, T. Lu, H. Kruse, H. Emamian, Exploring nature and predicting strength of hydrogen bonds: a correlation analysis between atoms-in-molecules descriptors, binding energies, and energy components of symmetry-adapted perturbation theory, J. Comput. Chem. 40 (2019) 2868–2881, doi:10.1002/jcc.26068.

[38] S.W. Zhu, H.P. Li, W.S. Zhu, W. Jiang, C. Wang, P.W. Wu, Q. Zhang, H.M. Li, Vibrational analysis and formation mechanism of typical deep eutectic solvents: an experimental and theoretical study, J. Mol. Graph. Model. 68 (2016) 158–175, doi:10.1016/j.jmgm.2016.05.003.

[39] S.L. Perkins, P. Painter, C.M. Colina, Experimental and computational studies of choline chloride-based deep eutectic solvents, J. Chem. Eng. Data. 59 (2014) 3652–3662, doi:10.1021/je500520h.

[40] S. Manzetti, T. Lu, The geometry and electronic structure of Aristolochic acid: Possible implications for a frozen resonance, J. Phys. Org. Chem. 26 (2013) 473–483, doi:10.1002/poc.3111.

[41] Z.Y. Liu, T. Lu, Q.X. Chen, Intermolecular interaction characteristics of the all-carboatomic ring, cyclo[18]carbon: Focusing on molecular adsorption and stacking, Carbon N. Y. 171 (2021) 514–523, doi:10.1016/j.carbon.2020.09.048.